

SFFedMC: Spurious-Free Federated Multi-View Clustering via Privacy-Preserving Confounder Balancing

Xianjie Guo

School of Computer Science, Nanjing
University of Posts and
Telecommunications
State Key Laboratory of Tibetan
Intelligence, Nanjing University of
Posts and Telecommunications
State Key Lab. of Novel Software
Technology, Nanjing University
Nanjing, Jangsu, China
xianjieguo@njupt.edu.cn

Kui Yu

School of Computer Science and
Information Engineering, Hefei
University of Technology
Hefei, Anhui, China
yukui@hfut.edu.cn

Shuai Yang

School of Artificial Intelligence,
Anhui Agricultural University
Hefei, Anhui, China
yangs@ahau.edu.cn

Xiaoli Tang

College of Computing and Data
Science, Nanyang Technological
University
Singapore
xiaoli001@e.ntu.edu.sg

Han Yu

College of Computing and Data
Science, Nanyang Technological
University
Singapore
han.yu@ntu.edu.sg

MingCai Chen*

School of Computer Science, Nanjing
University of Posts and
Telecommunications
State Key Lab. of Novel Software
Technology, Nanjing University
Nanjing, Jangsu, China
chenmc@njupt.edu.cn

Abstract

Federated multi-view clustering (FMVC) enables collaborative data grouping across distributed silos while preserving privacy. However, existing FMVC methods typically rely on statistical correlations between observed features and cluster labels, making them vulnerable to *spurious correlations* caused by hidden confounders (e.g., selection bias), which are often amplified by the non-IID nature of federated views. In this paper, we propose SFFedMC (Spurious-Free Federated Multi-View Clustering), the first framework to incorporate causal inference into FMVC. By leveraging a structural causal model (SCM), we identify the “backdoor paths” that mislead clustering. We then develop a privacy-preserving confounder balancing mechanism that learns optimal sample weights to minimize distributional discrepancies, effectively isolating causal features (e.g., essential semantic attributes) from non-causal noise without exchanging raw data. Integrated into a communication-efficient matrix factorization framework, SFFedMC eliminates spurious associations while maintaining low bandwidth overhead. Extensive experiments on eight real-world datasets demonstrate that SFFedMC significantly outperforms existing state-of-the-art baselines. The source code is available at <https://github.com/Xianjie-Guo/SFFedMC>.

*MingCai Chen is the corresponding author.



This work is licensed under a Creative Commons Attribution 4.0 International License.
MM '26, Rio de Janeiro, Brazil.

© 2026 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-2213-4/2026/11
<https://doi.org/10.1145/3767308.3836577>

CCS Concepts

• **Computing methodologies** → **Cluster analysis**; **Causal reasoning and diagnostics**; **Image representations**.

Keywords

Federated learning; Multi-view clustering; Causal inference

ACM Reference Format:

Xianjie Guo, Kui Yu, Shuai Yang, Xiaoli Tang, Han Yu, and MingCai Chen. 2026. SFFedMC: Spurious-Free Federated Multi-View Clustering via Privacy-Preserving Confounder Balancing. In *Proceedings of the 34th ACM International Conference on Multimedia (MM '26)*, November 10–14, 2026, Rio de Janeiro, Brazil. ACM, New York, NY, USA, 9 pages. <https://doi.org/10.1145/3767308.3836577>

1 Introduction

Background. Multi-view clustering (MVC) has emerged as a fundamental unsupervised learning paradigm, designed to group data samples by integrating complementary information from distinct representations or views [20, 22, 23, 34–37]. In conventional centralized settings, MVC requires collecting all views into a single repository. However, in the era of big data, real-world applications such as healthcare, finance, and cross-organization recommendation systems often face the “data island” dilemma [12, 27, 28]. Due to increasingly stringent data privacy regulations (e.g., GDPR, HIPAA) and commercial competition [26], data views are often distributed across isolated clients (silos) and cannot be physically aggregated. To address this challenge, Federated multi-view clustering (FMVC) [19, 21, 31] was proposed, which enables collaborative clustering across distributed clients without exchanging raw data,

thereby bridging the gap between data utilization and privacy protection. For instance, in medical diagnosis, a patient’s MRI scans, CT images, and genetic records may be stored in different hospitals. FMVC allows these institutions to jointly identify patient subtypes (clusters) while keeping sensitive medical records locally.

Related Work. Existing FMVC studies have made significant strides in preserving privacy while maintaining clustering performance. Due to space limitations, we provide a brief overview here and a comprehensive discussion in Appendix D¹. Generally, these methods can be categorized based on the data partitioning strategies: (1) *Horizontal FMVC*: Methods like FedCM, FedFCM [1], FedMVFC [15], and Fed-MVKM [31] address scenarios where clients share the same feature space but hold different samples. They typically aggregate local gradients or prototypes to update a global model. (2) *Vertical FMVC*: This is the more common scenario in MVC where clients share sample IDs but hold different feature views. Prominent works include CeFMC [19], FedMVL [16], FMVC-IMK [9], FMVFCMSP [8], SFOMVC-TR [7], TensorFMVC [10], and ESFMC [3]. These methods often utilize vertical federated learning (VFL) to align representations or consensus matrices via encrypted entity alignment. Notably, ESFMC leverages information-theoretic feature splitting to share only task-relevant features, enhancing privacy protection. (3) *Heterogeneous/Incomplete FMVC*: Addressing more complex real-world constraints, recent works like FedDMVC [5], FIM-GNNs [30], FMCSC [4], and FCUIF [25] tackle issues of missing views, unaligned data, or heterogeneous client architectures through techniques like data imputation, adaptive alignment, or graph neural networks.

Motivation. Although existing FMVC approaches have achieved remarkable success, they predominantly focus on capturing the statistical correlation between features and cluster labels. For example, the state-of-the-art method CeFMC [19] efficiently approximates representations using pseudo-labels and centroids. However, these methods rely on a strong assumption: *all observed feature-label correlations are reliable for clustering*. This assumption ignores the pervasive issue of **spurious correlations** caused by confounders. Inspired by the causal perspective in multi-view learning [32], we recognize that view features can be theoretically decomposed into:

- (1) **Causal Features**: Features that directly determine the semantic label (e.g., the shape of an animal’s face).
- (2) **Non-causal Features**: Features irrelevant to the semantic content (e.g., the background grass in an image).
- (3) **Confounders**: Hidden variables (e.g., selection bias in data collection) that influence both the features and the labels.

As illustrated in Figure 1, confounders can create a “backdoor path”, causing non-causal features to exhibit strong but spurious statistical correlations with cluster labels [13]. In a federated setting, this risk is amplified because non-IID distributions across clients can introduce severe selection biases [11, 14]. Standard FMVC algorithms, which blindly maximize feature-label associations, may be misled by these confounders, resulting in clusters dominated by background noise rather than semantic truth. Therefore, eliminating spurious correlations in a privacy-preserving manner is a critical yet unexplored frontier in FMVC.

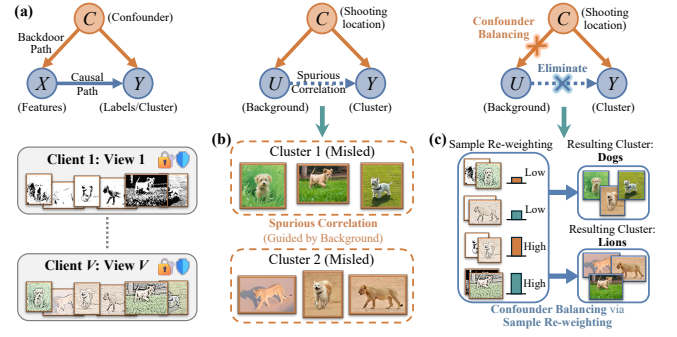


Figure 1: The motivation of SFFedMC. (a) The Structural Causal Model (SCM) reveals that the Confounder C (e.g., background bias) creates a spurious link between non-causal features U and labels Y . (b) Traditional FMVC fails to distinguish causal from non-causal features due to this spurious correlation. (c) Our SFFedMC re-weights samples to balance the confounder distribution, blocking the backdoor path and recovering true semantic clusters.

Contributions. To bridge this gap, we propose a novel algorithm named SFFedMC (Spurious-Free Federated Multi-View Clustering). SFFedMC is designed to perform causality-aware clustering by effectively cutting off backdoor paths in a federated environment. Our main contributions are summarized as follows:

- We present the first attempt to incorporate causal inference into federated multi-view clustering. By identifying and mitigating the impact of confounders, we tackle the bottleneck of spurious correlations that limits existing FMVC performance.
- We propose a privacy-preserving confounder balancing mechanism. By learning optimal sample weights that minimize the distributional discrepancy between treated and control groups across views, we virtually simulate a randomized experiment to eliminate spurious associations without sharing raw data.
- We design a communication-efficient optimization framework. SFFedMC inherits the low-bandwidth advantages of matrix-factorization-based VFL while integrating the causal regularization term, ensuring high efficiency suitable for edge computing scenarios.
- Extensive experiments demonstrate that SFFedMC significantly outperforms state-of-the-art federated baselines, validating the effectiveness of our causal-driven approach in uncovering true cluster structures.

2 Preliminaries

In this section, we formally define the problem of federated multi-view clustering (FMVC) and introduce the structural causal model (SCM) to characterize the spurious correlations. For clarity, a summary of the main notations used throughout this paper is provided in Table 2 of Appendix A.

2.1 Problem Formulation

We consider a federated system consisting of V clients and a central server. Let $\mathcal{V} = \{1, \dots, V\}$ denote the set of views, and $\mathcal{D} = \{\mathbf{X}^{(v)}\}_{v=1}^V$ denote the distributed multi-view dataset, where $\mathbf{X}^{(v)} =$

¹Appendices are available at <https://xianjie-guo.github.io/pdf/SFFedMC-app.pdf>.

$[\mathbf{x}_1^{(v)}, \dots, \mathbf{x}_N^{(v)}]^\top \in \mathbb{R}^{N \times D_v}$ is the feature matrix held by the v -th client. Here, N is the number of samples aligned across views, and D_v is the feature dimension of the v -th view. The samples are vertically partitioned, meaning clients share the sample ID space but hold disjoint feature spaces.

The objective of FMVC is to partition the N samples into K disjoint clusters $C = \{C_1, \dots, C_K\}$ by learning a consensus partition matrix $\mathbf{y} \in \{0, 1\}^{N \times K}$ such that $y_{ik} = 1$ if sample i belongs to cluster k , and 0 otherwise. This must be achieved under the strict privacy constraint that raw data $\mathbf{X}^{(v)}$ never leaves the local client.

2.2 Federated Multi-view Hilbert Space

To capture non-linear local geometries, we employ kernel mapping. Let $\phi_v : \mathbb{R}^{D_v} \rightarrow \mathcal{H}^{(v)}$ be a feature map to a Reproducing Kernel Hilbert Space (RKHS) [2] $\mathcal{H}^{(v)}$. The kernel function is defined as $\mathcal{K}_v(\mathbf{x}_i^{(v)}, \mathbf{x}_j^{(v)}) = \langle \phi_v(\mathbf{x}_i^{(v)}), \phi_v(\mathbf{x}_j^{(v)}) \rangle_{\mathcal{H}^{(v)}}$, which induces a semi-positive definite (SPD) kernel matrix $\mathbf{K}^{(v)} \in \mathbb{S}_+^{N \times N}$. The global objective is to find a set of orthogonal latent representations $\mathbf{H}^{(v)}$ such that:

$$\mathbf{H}^{(v)} \in \mathcal{S}\mathbb{L}(N, c) = \{\mathbf{H} \in \mathbb{R}^{N \times c} \mid \mathbf{H}^\top \mathbf{H} = \mathbf{I}_c\}, \quad (1)$$

where $\mathcal{S}\mathbb{L}(N, c)$ denotes the Stiefel manifold, ensuring that the learned features are non-redundant and scaled.

2.3 Causal Structural Modeling of Spuriousness

To address spurious correlations, we formulate the data generation process using an SCM $\mathfrak{M} = \langle \mathbf{U}, \mathbf{V}, \mathcal{F}, P(\mathbf{U}) \rangle$. For a specific view v , the variables are defined as follows:

- Y : The semantic cluster label (Effect).
- $\mathbf{S}^{(v)}$: Causal features (e.g., shape) that directly determine Y .
- $\mathbf{N}^{(v)}$: Non-causal features (e.g., background) that should be independent of Y .
- C : A latent confounder (e.g., selection bias) affecting both $\mathbf{N}^{(v)}$ and Y .

The following structural equations govern the causal relationships:

$$\begin{aligned} C &:= \epsilon_C, \quad \mathbf{S}^{(v)} := f_S(\epsilon_S), \quad Y := f_Y(\mathbf{S}^{(v)}, C, \epsilon_Y), \\ \mathbf{N}^{(v)} &:= f_N(C, \epsilon_N), \quad \mathbf{X}^{(v)} := \phi(\mathbf{S}^{(v)}, \mathbf{N}^{(v)}), \end{aligned} \quad (2)$$

where ϵ denotes independent exogenous noise.

Spurious Correlation. The path $Y \leftarrow C \rightarrow \mathbf{N}^{(v)}$ forms a “backdoor path” (confounding arc). Consequently, the conditional probability $P(Y|\mathbf{N}^{(v)}) \neq P(Y)$ creates a statistical dependency between the label Y and non-causal features $\mathbf{N}^{(v)}$. Standard clustering algorithms that maximize likelihood $P(\mathbf{X}^{(v)}|Y)$ will inevitably exploit this spurious correlation, leading to bias.

Backdoor Adjustment. To recover the true causal effect, we must perform an intervention $do(X)$. According to Pearl’s backdoor criterion [24], if we can condition on the confounder C , the spurious path is blocked:

$$P(Y|do(X)) = \sum_c P(Y|X, C=c)P(C=c). \quad (3)$$

Since C is latent, we propose to approximate this adjustment via sample re-weighting [17]. Specifically, we seek a re-weighted density P_τ such that the Kullback-Leibler (KL) divergence between the conditional and marginal distributions of the confounder vanishes:

$$\mathbb{D}_{KL}(P_\tau(C | \mathbf{N}^{(v)}) \| P_\tau(C)) \rightarrow 0. \quad (4)$$

In our framework, the de-confounding is operationalized via the maximum mean discrepancy (MMD) [18] in the RKHS $\mathcal{H}^{(v)}$:

3 The Proposed SFFedMC Method

As shown in Figure 2, we propose SFFedMC, which integrates causal de-confounding into a federated matrix factorization framework. The detailed pseudocode of SFFedMC is provided in Algorithm 1 of Appendix E.

3.1 Privacy-Preserving Confounder Balancing

The core innovation of SFFedMC is the derivation of sample-specific causal weights $\tau \in \Delta^{N-1}$ to neutralize the influence of the confounder C , where Δ^{N-1} is the probability simplex.

Treatment Assignment Mechanism. Since the confounder C is latent, we utilize the heterogeneity in the learned representations as a proxy. For each view v , given the current latent representation $\mathbf{H}^{(v)} \in \mathbb{R}^{N \times c}$, we define a binary treatment assignment indicator $A_v(i) \in \{0, 1\}$ based on the dominant principal component:

$$A_v(i) = \mathbb{I}([\mathbf{H}^{(v)}]_{i,\star} \geq \text{median}([\mathbf{H}^{(v)}]_{\cdot,\star})), \quad (5)$$

where index $\star$ denotes the dimension with maximum variance. By utilizing Eq. (5), we can partition the dataset into a treatment group $\mathcal{T}_v = \{i | A_v(i) = 1\}$ and a control group $\mathcal{C}_v = \{i | A_v(i) = 0\}$.

Causal Balancing via MMD. To eliminate the dependency on C , we require the distribution of representations in the weighted population to be invariant across groups. We adopt the maximum mean discrepancy (MMD) [18] in the RKHS $\mathcal{H}^{(v)}$ as the discrepancy metric. The causal regularization loss for view v is formulated as:

$$\mathcal{L}_{\text{causal}}^{(v)}(\tau) = \|\mu_{\mathcal{T}_v}(\tau) - \mu_{\mathcal{C}_v}(\tau)\|_{\mathcal{H}^{(v)}}^2, \quad (6)$$

where $\mu_{\mathcal{T}_v}$ and $\mu_{\mathcal{C}_v}$ are the weighted mean embeddings:

$$\mu_{\mathcal{T}_v}(\tau) = \frac{(\mathbf{H}^{(v)})^\top (\tau \odot \mathbf{a}_v)}{\mathbf{1}^\top (\tau \odot \mathbf{a}_v)}, \quad \mu_{\mathcal{C}_v}(\tau) = \frac{(\mathbf{H}^{(v)})^\top (\tau \odot (\mathbf{1} - \mathbf{a}_v))}{\mathbf{1}^\top (\tau \odot (\mathbf{1} - \mathbf{a}_v))}. \quad (7)$$

Here, $\odot$ denotes the Hadamard product, and $\mathbf{a}_v$ is the vector form of A_v . Minimizing this term (i.e., Eq. (6)) forces the moments of the distributions to match, effectively satisfying the *covariate balance* condition $P_\tau(\mathbf{H}|\mathbf{A}=1) \approx P_\tau(\mathbf{H}|\mathbf{A}=0)$, thereby blocking the backdoor path.

3.2 Global Objective Function

SFFedMC formulates the clustering problem as a weighted non-negative matrix factorization constrained by the causal regularizer. Specifically, SFFedMC unifies kernel-based representation learning, consensus clustering, and causal balancing into a single constrained

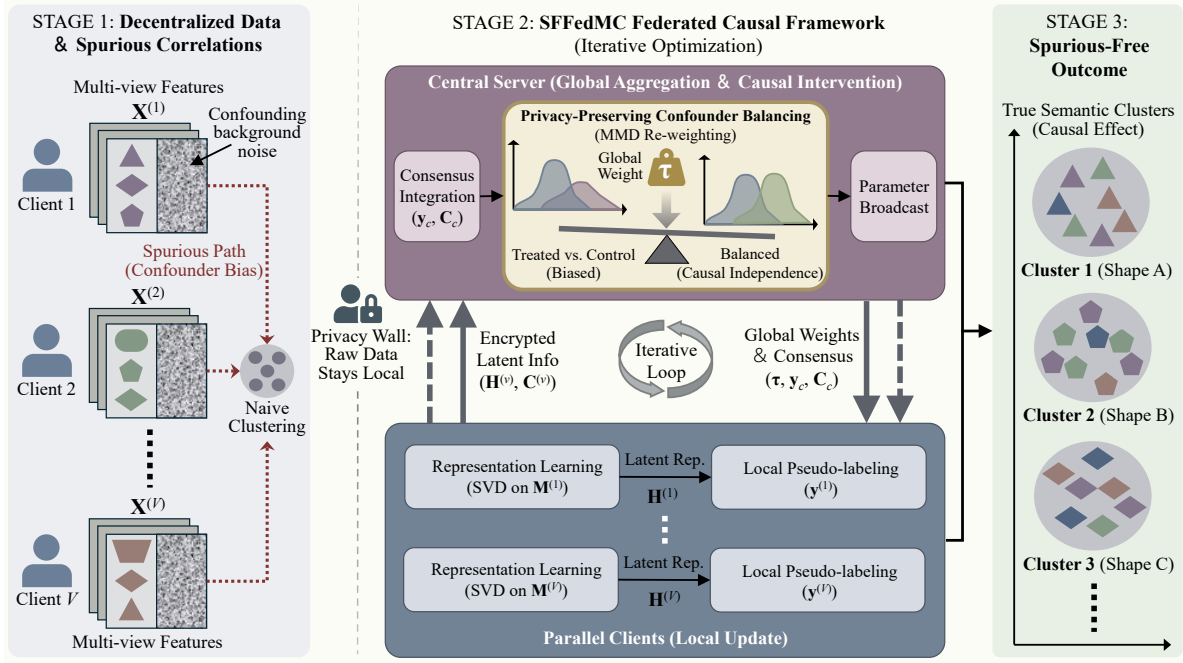


Figure 2: The framework of our proposed SFFedMC method.

optimization problem. The global objective function $\mathcal{J}(\Theta)$ is defined as:

$$\max_{\Theta} \sum_{v=1}^V \left[\underbrace{\text{Tr} \left((\mathbf{H}^{(v)})^T \Gamma(\tau) \mathbf{K}^{(v)} \mathbf{H}^{(v)} \right)}_{\mathcal{J}_{\text{align}}: \text{Weighted Kernel Alignment}} + \lambda \underbrace{\text{Tr} \left((\mathbf{H}^{(v)})^T \mathbf{Y}^{(v)} \mathbf{C}^{(v)} \right)}_{\mathcal{J}_{\text{local}}: \text{Local Approximation}} \right] + \beta \underbrace{\text{Tr} \left(\mathbf{H}_c^T \mathbf{Y}_c \mathbf{C}_c \right)}_{\mathcal{J}_{\text{global}}: \text{Global Consensus}} - \gamma \sum_{v=1}^V \mathcal{L}_{\text{causal}}^{(v)}(\tau), \quad (8)$$

subject to the following orthogonal and probabilistic constraints:

$$\text{s.t. } (\mathbf{H}^{(v)})^T \mathbf{H}^{(v)} = \mathbf{I}_c, \quad \mathbf{C}^{(v)} (\mathbf{C}^{(v)})^T = \mathbf{I}_c, \quad \mathbf{C}_c \mathbf{C}_c^T = \mathbf{I}_c, \\ \forall v \in \mathcal{V}, \quad \tau \in \Delta^{N-1} = \{\tau \in \mathbb{R}^N \mid \tau \geq \mathbf{0}, \tau^T \mathbf{1} = 1\}. \quad (9)$$

Here, $\Theta = \{ \{ \mathbf{H}^{(v)}, \mathbf{Y}^{(v)}, \mathbf{C}^{(v)} \}_{v=1}^V, \mathbf{Y}_c, \mathbf{C}_c, \tau \}$ is the set of parameters. The matrix $\Gamma(\tau) = \text{diag}(\tau)$ encodes the causal weights and introduces the causal weights into the kernel alignment, ensuring that the learned subspace prioritizes samples that are robust to confounding bias. $\mathbf{K}^{(v)} = \mathbf{X}^{(v)} (\mathbf{X}^{(v)})^T$ is the local kernel matrix. $\mathbf{Y}^{(v)}$ and $\mathbf{Y}_c$ are local and global pseudo-label matrices, respectively.

3.3 Optimization Strategy

The objective in Eq. (8) couples non-convex matrix factorization with manifold constraints and a causal balancing regularizer. Therefore, we develop a *Block Coordinate Ascent with Causal Re-weighting* (BCA-CR) algorithm that alternates between client-side representation learning on Stiefel manifolds and server-side consensus aggregation with MMD-based confounder balancing.

3.3.1 Client Update: Orthogonal Procrustes on Stiefel Manifold.

Representation Subproblem. For each view v , fixing $\{ \mathbf{Y}^{(v)}, \mathbf{C}^{(v)}, \mathbf{C}_c, \mathbf{Y}_c, \tau \}$, we solve:

$$\max_{\mathbf{H}^{(v)} \in \mathcal{S}_{\sqcup}(N, c)} \text{Tr} \left[(\mathbf{H}^{(v)})^T \Gamma(\tau) \mathbf{K}^{(v)} \mathbf{H}^{(v)} \right] + \lambda \text{Tr} \left[(\mathbf{H}^{(v)})^T \mathbf{Y}^{(v)} \mathbf{C}^{(v)} \right], \quad (10)$$

where $\mathcal{S}_{\sqcup}(N, c) = \{ \mathbf{H} \in \mathbb{R}^{N \times c} : \mathbf{H}^T \mathbf{H} = \mathbf{I}_c \}$ is the Stiefel manifold, $\Gamma(\tau) = \text{diag}(\tau)$, and $\mathbf{K}^{(v)} = \mathbf{X}^{(v)} (\mathbf{X}^{(v)})^T$. Then, we construct the gradient-based auxiliary matrix:

$$\mathbf{M}^{(v)} = \Gamma(\tau) \mathbf{K}^{(v)} \mathbf{H}^{(v)} + \lambda \mathbf{Y}^{(v)} \mathbf{C}^{(v)} \in \mathbb{R}^{N \times c}. \quad (11)$$

The solution to Eq. (10) is the Procrustes projection:

$$\mathbf{H}^{(v)} = \mathbf{U}_v \mathbf{V}_v^T, \quad \text{where } \mathbf{M}^{(v)} = \mathbf{U}_v \Sigma_v \mathbf{V}_v^T. \quad (12)$$

This updates $\mathbf{H}^{(v)}$ via compact SVD of $\mathbf{M}^{(v)}$ in $O(Nc^2)$ time.

Label Assignment. Given updated $\mathbf{H}^{(v)}$, local pseudo-labels are:

$$\mathbf{y}_i^{(v)} = \arg \max_{k \in [K]} \left\{ \lambda \mathbf{H}_{i,:}^{(v)} (\mathbf{C}_{k,:}^{(v)})^T + \beta [\mathbf{C}_c]_{y_c(i),:} (\mathbf{C}_{k,:}^{(v)})^T \right\}. \quad (13)$$

Local centroids are then orthonormalized:

$$\mathbf{C}^{(v)} = \mathbf{V}_v \mathbf{U}_v^T, \quad \text{where } (\mathbf{Y}^{(v)})^T \mathbf{H}^{(v)} = \mathbf{U}_v \Sigma_v \mathbf{V}_v^T. \quad (14)$$

Each client uploads $(\mathbf{Y}^{(v)}, \mathbf{C}^{(v)})$ to the server, totaling $O(N + Kc)$ scalars per view.

3.3.2 Server Update: Consensus Aggregation and Causal Balancing.

Global Centroid. The central server forms $\mathbf{H}_c = [\mathbf{Y}^{(1)} \mathbf{C}^{(1)}, \dots, \mathbf{Y}^{(V)} \mathbf{C}^{(V)}] \in \mathbb{R}^{N \times \sum c_v}$. Solving:

$$\max_{\mathbf{C}_c \mathbf{C}_c^T = \mathbf{I}_K} \text{Tr}[(\mathbf{C}_c)^T (\mathbf{Y}_c)^T \mathbf{H}_c] \Rightarrow \mathbf{C}_c = \mathbf{V} \mathbf{U}^T, \quad (15)$$

where $\mathbf{H}_c^T \mathbf{Y}_c = \mathbf{U} \Sigma \mathbf{V}^T$ is the reduced SVD. Numerically scale: $\mathbf{C}_c \leftarrow (K/N) \mathbf{C}_c$.

Weighted Consensus Labels. Instead of uniform voting, we use causal weights:

$$(\mathbf{y}_c)_i = \arg \max_{k \in [K]} \{\tau_i \cdot [\mathbf{H}_c \mathbf{C}_c^\top]_{ik}\}. \quad (16)$$

Causal Weight Update via Projected Gradient Descent. The core innovation is minimizing the multi-view MMD:

$$\min_{\boldsymbol{\tau} \in \Delta^{N-1}} \mathcal{L}_{\text{causal}}(\boldsymbol{\tau}) = \sum_{v=1}^V \|\boldsymbol{\mu}_{\mathcal{T}_v}(\boldsymbol{\tau}) - \boldsymbol{\mu}_{\mathcal{C}_v}(\boldsymbol{\tau})\|_{\mathcal{H}^{(v)}}^2, \quad (17)$$

where $\Delta^{N-1} = \{\boldsymbol{\tau} \in \mathbb{R}_+^N : \sum_i \tau_i = 1\}$ and the weighted kernel means are:

$$\boldsymbol{\mu}_{\mathcal{T}_v}(\boldsymbol{\tau}) = \frac{(\mathbf{H}^{(v)})^\top (\boldsymbol{\tau} \odot \mathbf{a}_v)}{\mathbf{1}^\top (\boldsymbol{\tau} \odot \mathbf{a}_v)}, \quad \boldsymbol{\mu}_{\mathcal{C}_v}(\boldsymbol{\tau}) = \frac{(\mathbf{H}^{(v)})^\top (\boldsymbol{\tau} \odot (\mathbf{1} - \mathbf{a}_v))}{\mathbf{1}^\top (\boldsymbol{\tau} \odot (\mathbf{1} - \mathbf{a}_v))}, \quad (18)$$

with $\mathbf{a}_v \in \{0, 1\}^N$ indicating treatment assignment (see Eq. (5) for details). Expanding the squared RKHS norm:

$$\begin{aligned} \|\boldsymbol{\mu}_{\mathcal{T}} - \boldsymbol{\mu}_{\mathcal{C}}\|_{\mathcal{H}}^2 &= \|\boldsymbol{\mu}_{\mathcal{T}}\|^2 + \|\boldsymbol{\mu}_{\mathcal{C}}\|^2 - 2\langle \boldsymbol{\mu}_{\mathcal{T}}, \boldsymbol{\mu}_{\mathcal{C}} \rangle \\ &= \frac{\|(\mathbf{H}^{(v)})^\top (\boldsymbol{\tau} \odot \mathbf{a})\|_2^2}{(\mathbf{1}^\top (\boldsymbol{\tau} \odot \mathbf{a}))^2} + \frac{\|(\mathbf{H}^{(v)})^\top (\boldsymbol{\tau} \odot (\mathbf{1} - \mathbf{a}))\|_2^2}{(\mathbf{1}^\top (\boldsymbol{\tau} \odot (\mathbf{1} - \mathbf{a})))^2} \\ &\quad - 2 \frac{(\boldsymbol{\tau} \odot \mathbf{a})^\top \mathbf{H}^{(v)} (\mathbf{H}^{(v)})^\top (\boldsymbol{\tau} \odot (\mathbf{1} - \mathbf{a}))}{(\mathbf{1}^\top (\boldsymbol{\tau} \odot \mathbf{a}))(\mathbf{1}^\top (\boldsymbol{\tau} \odot (\mathbf{1} - \mathbf{a})))}. \end{aligned} \quad (19)$$

The gradient $\nabla_{\boldsymbol{\tau}} \mathcal{L}_{\text{causal}}^{(v)}$ (derivation in [Appendix C](#)) combines quotient rule terms. For $i \in \mathcal{T}_v$:

$$\frac{\partial \mathcal{L}_{\text{causal}}^{(v)}}{\partial \tau_i} = \frac{2}{Z_{\mathcal{T}}} [(\boldsymbol{\mu}_{\mathcal{T}} - \boldsymbol{\mu}_{\mathcal{C}})^\top (\mathbf{h}_i^{(v)} - \boldsymbol{\mu}_{\mathcal{T}})] + O(Z_{\mathcal{T}}^{-2}), \quad (20)$$

where $\mathbf{h}_i \in \mathbb{R}^c$ is the latent representation of sample i , and $Z_{\mathcal{T}} = \sum_{j \in \mathcal{T}_v} \tau_j$. Aggregating over all views:

$$\mathbf{g} = \sum_{v=1}^V \nabla_{\boldsymbol{\tau}} \mathcal{L}_{\text{causal}}^{(v)}(\boldsymbol{\tau}). \quad (21)$$

Update via projected gradient descent:

$$\boldsymbol{\tau}^{(t+1)} = \Pi_{\Delta}(\boldsymbol{\tau}^{(t)} - \eta \mathbf{g}), \quad (22)$$

where Π_{Δ} is Duchi's $O(N \log N)$ simplex projection [6], and η is the learning rate for the projected gradient descent on $\boldsymbol{\tau}$. The server broadcasts $(\mathbf{y}_c, \mathbf{C}_c, \boldsymbol{\tau})$ to clients, requiring $O(N + K \sum c_v)$ scalars.

3.4 Theoretical Analysis

In this section, we establish the theoretical foundations of SFFedMC, focusing on three key aspects: the *identifiability of causal clusters*, *convergence*, and *privacy guarantees*. These theoretical results ensure that our method is not only empirically effective but also mathematically sound.

First, we address the core motivation of this work: eliminating spurious correlations. A crucial question is whether the proposed re-weighting strategy effectively isolates the causal features. Theorem 3.1 provides an affirmative answer by linking the vanishing MMD loss to the independence between non-causal features and cluster labels.

Theorem 3.1. *Let $\mathcal{S}$ and $\mathcal{N}$ denote the sets of causal and non-causal features, respectively. Assume the Causal Sufficiency [33] holds. If the global weight vector $\boldsymbol{\tau}^*$ minimizes the causal regularizer such that*

$\sum_v \mathcal{L}_{\text{causal}}^{(v)}(\boldsymbol{\tau}^) = 0$, then under the weighted distribution $P_{\boldsymbol{\tau}^*}$, the spurious correlation path is blocked. Specifically, the mutual information between $\mathcal{N}$ and the labels vanishes: $I(\mathcal{N}; Y | \boldsymbol{\tau}^*) = 0$. Consequently, the optimization of $\mathbf{H}$ relies solely on the causal features $\mathcal{S}$.*

Theorem 3.1 serves as the cornerstone of our framework, justifying the use of MMD minimization as a proxy for causal intervention. Next, we consider the stability of the training process. Since the objective function in Eq. (8) involves non-convex matrix factorization coupled with a causal regularizer, establishing convergence is non-trivial. Theorem 3.2 guarantees that our alternating optimization strategy reliably reaches a stationary point.

Theorem 3.2 (Convergence). *The objective function $\mathcal{J}(\Theta)$ in Eq. (8) is upper-bounded. The alternating optimization algorithm of SFFedMC generates a sequence of parameter states $\{\Theta^{(t)}\}_{t=1}^{\infty}$ such that the objective value is monotonically non-decreasing (i.e., $\mathcal{J}(\Theta^{(t+1)}) \geq \mathcal{J}(\Theta^{(t)})$). The sequence converges to a local stationary point Θ^* .*

Finally, since SFFedMC operates in a federated setting, we must quantify its privacy properties. Lemma 3.3 formally states that the raw data and local topological structures remain undisclosed.

Lemma 3.3 (Privacy Preservation). *During the execution of SFFedMC, the raw data matrices $\mathbf{X}^{(v)}$ and the local kernel matrices $\mathbf{K}^{(v)}$ are strictly confined to the local clients. The information exchanged between clients and the server is limited to the low-rank latent representations $\mathbf{H}^{(v)}$, centroids $\mathbf{C}^{(v)}$, and scalar weights $\boldsymbol{\tau}$, preventing the reconstruction of raw features.*

The detailed proofs for Theorem 3.1, Theorem 3.2, and Lemma 3.3 are provided in [Appendix B.1](#), [Appendix B.2](#), and [Appendix B.3](#), respectively.

3.5 Communication Cost Analysis

Efficiency is a critical constraint in a federated setting, where communication overhead often becomes the bottleneck. In this section, we provide an analysis of the communication complexity of SFFedMC, highlighting the minimal overhead introduced by our causal intervention mechanism.

Per-Iteration Communication Breakdown. In each iteration t , the communication process involves two phases:

- (1) **Uplink (Client $\rightarrow$ Server):** Each of the V clients transmits its local pseudo-labels $\mathbf{y}^{(v)} \in \mathbb{R}^N$ (represented efficiently as integer indices) and the local centroid matrix $\mathbf{C}^{(v)} \in \mathbb{R}^{K \times c}$. The payload size for each client is $O(N \log K + K \cdot c)$ bits, where the logarithmic factor accounts for the entropy encoding of label indices.
- (2) **Downlink (Server $\rightarrow$ Client):** The server broadcasts the global consensus labels $\mathbf{y}_c \in \mathbb{R}^N$, the global centroids $\mathbf{C}_c \in \mathbb{R}^{K \times c}$, and the updated sample weights $\boldsymbol{\tau} \in \mathbb{R}^N$. The broadcast payload size is $O(N \log K + K \cdot c + N)$ bits per client.

Aggregating over T iterations and V clients, the total communication cost is:

$$C_{\text{total}} = O(T \cdot V \cdot (N + K \cdot c)). \quad (23)$$

Causal Module Overhead. The overhead introduced by the causal intervention mechanism is minimal and consists solely of transmitting the weight vector $\boldsymbol{\tau}$ (size N) from the server to clients. This

represents only $\frac{N}{N+K \cdot c} \approx \frac{1}{1+K \cdot c/N}$ of the total downlink payload. In typical settings where $K \cdot c \ll N$, this fraction is negligible. Importantly, the causal balancing computation (MMD minimization and gradient projection) occurs entirely on the server side and does not introduce any additional client-to-server communication, preserving the communication efficiency of the baseline matrix factorization framework.

Scalability Analysis. Eq. (23) indicates that the communication complexity of SFFedMC is *independent of the feature dimension D_v* . The dimension-independent nature ensures favorable scaling properties. As the number of samples N or the number of views V increases, the communication cost grows linearly, which is optimal for federated scenarios.

4 Experimental Evaluation

4.1 Experimental Settings

Datasets. We evaluate on eight real-world multi-view datasets covering image and text domains: Caltech101-7 (6 views, 7 classes), Hdigit (2 views, 10 classes), Mfeat (6 views, 10 classes), Multi-Fashion (3 views, 10 classes), MSRC-v1 (4 views, 7 classes), ORL (3 views, 40 classes), WebKB (3 views, 4 classes), and Yale (3 views, 15 classes). Detailed dataset statistics are provided in [Appendix F.1](#).

Comparison Baselines. We compare against seven state-of-the-art FMVC methods: (1) **FedMVL** [16]: addresses feature heterogeneity across distributed nodes with $L_{2,1}$ regularization and fault tolerance mechanisms; (2) **FedCMv** [1]: lossless vertical federated c-means that preserves exact cluster assignments as centralized execution; (3) **FedFCMv** [1]: lossless vertical federated fuzzy c-means with privacy-preserving soft clustering; (4) **CeFMC** [19]: communication-efficient framework using pseudo-label and centroid sharing with linear kernel approximation; (5) **FMVFCMSP** [8]: federated fuzzy c-means clustering with tensor Schatten- p norm regularization to exploit inter-view complementary information; (6) **ESFMC** [3]: information-theoretic feature splitting method that shares only task-relevant features to balance privacy and performance; (7) **FCUIF** [25]: a federated method that handles unaligned and incomplete multi-view data via adaptive alignment matrices and unsupervised imputation quality refinement.

Evaluation Metrics. We employ four clustering metrics: Clustering Accuracy (ACC), Normalized Mutual Information (NMI), F-score, and Adjusted Rand Index (ARI). All metrics are with higher values indicating better performance.

Implementation Details. For all methods, we perform 5 independent runs with different random seeds and report the mean $\pm$ standard deviation. Hyperparameters are set as: $\lambda = 1.0$, $\beta = 1.0$, $\gamma = 0.01$, and $T_{\max} = 100$. Additional implementation details are provided in [Appendix F.2](#).

4.2 Clustering Performance

Table 1 summarizes the clustering performance across all datasets. SFFedMC achieves the best ACC on all datasets, demonstrating its superior ability to discover accurate cluster structures. Beyond ACC, SFFedMC also leads on NMI, F-score, and ARI in the majority of cases,

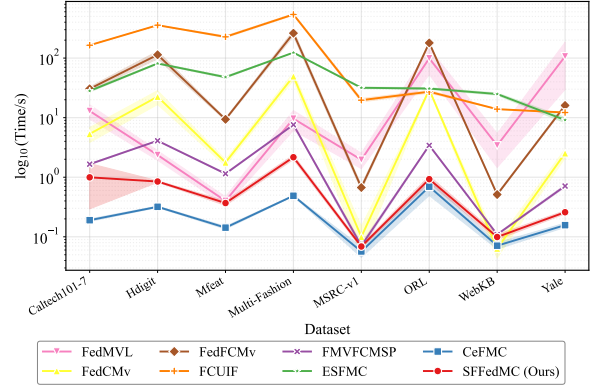


Figure 3: Runtime comparison across datasets.

reflecting consistently well-separated and compact cluster structures. On datasets where spurious correlations are prominent (such as Caltech101-7, Yale, ORL, and WebKB), the improvements are particularly pronounced, confirming the effectiveness of the causal confounder balancing mechanism. Notably, on Multi-Fashion, while SFFedMC dominates in ACC, it shows moderate performance on NMI, F-score, and ARI. We attribute this to the dataset’s inherent symmetry (three identical views of fashion items), where spurious correlations are less pronounced, reducing the advantage of causal regularization. Overall, the consistent gains across diverse datasets and metrics validate that SFFedMC learns more causally grounded representations than correlation-based baselines.

Figure 3 compares the computational efficiency of all methods. SFFedMC ranks second only to CeFMC in runtime across datasets, with only a marginal gap between them. Remarkably, on MSRC-v1, ORL, WebKB, and Yale, SFFedMC achieves nearly identical speed to CeFMC while delivering substantially superior clustering quality, demonstrating that the additional causal weight update step (Eq. (17)) incurs negligible computational overhead. In contrast, FCUIF and FedFCMv are consistently among the slowest methods overall, as their respective alignment-imputation and fuzzy membership optimization procedures introduce considerable per-iteration cost that compounds across datasets. This efficiency advantage makes SFFedMC well-suited for practical federated deployments where both accuracy and computational budget matter.

4.3 Ablation Study

To understand the contribution of each component, we design three ablations: (1) **SFFedMC-NoCR**: Remove causal regularization ($\gamma = 0$); (2) **SFFedMC-NoWK**: Replace weighted kernel with standard kernel; (3) **SFFedMC-NoCI**: Remove consensus integration ($\beta = 0$). Figure 4 shows that the full SFFedMC consistently outperforms all variants, confirming each component is indispensable. Removing causal regularization (NoCR) yields the largest degradation on datasets with strong spurious correlations (e.g., Caltech101-7, Yale, and ORL), while removing the weighted kernel (NoWK) is most harmful on multi-modal datasets such as Hdigit and WebKB. Removing consensus integration (NoCI) consistently hurts performance on datasets with fine-grained class structures. Results for NMI, F-score, and ARI are provided in [Appendix F.3](#).

Table 1: Clustering performance comparison on eight multi-view datasets. The best results are highlighted in bold and the second best are underlined. Results are reported as mean $\pm$ std (%) over 5 runs.

Method	Caltech101-7				Hdigit			
	ACC ($\uparrow$)	NMI ($\uparrow$)	F-score ($\uparrow$)	ARI ($\uparrow$)	ACC ($\uparrow$)	NMI ($\uparrow$)	F-score ($\uparrow$)	ARI ($\uparrow$)
FedMVL [16]	47.80 $\pm$ 1.51	54.39 $\pm$ 1.24	51.20 $\pm$ 1.06	35.37 $\pm$ 0.75	84.25 $\pm$ 2.20	80.56 $\pm$ 0.58	78.43 $\pm$ 0.25	75.99 $\pm$ 0.32
FedCMv [1]	48.28 $\pm$ 5.72	55.15 $\pm$ 3.74	53.67 $\pm$ 3.93	39.16 $\pm$ 4.41	72.41 $\pm$ 3.78	77.69 $\pm$ 2.66	70.44 $\pm$ 3.67	66.94 $\pm$ 4.07
FedFCMv [1]	45.07 $\pm$ 0.17	56.13 $\pm$ 0.52	52.75 $\pm$ 0.07	38.36 $\pm$ 0.18	84.95 $\pm$ 5.20	84.25 $\pm$ 2.15	81.71 $\pm$ 3.94	79.62 $\pm$ 4.41
FCUIF [25]	47.75 $\pm$ 1.07	55.68 $\pm$ 1.61	53.07 $\pm$ 0.38	38.87 $\pm$ 0.62	83.39 $\pm$ 6.23	79.07 $\pm$ 3.49	76.33 $\pm$ 5.73	73.65 $\pm$ 6.42
FMVFCMSP [8]	<u>50.84</u> $\pm$ 8.62	42.43 $\pm$ 4.56	54.56 $\pm$ 10.17	37.64 $\pm$ 10.73	43.42 $\pm$ 3.23	38.64 $\pm$ 1.62	34.02 $\pm$ 2.41	25.97 $\pm$ 2.81
ESFMC [3]	49.66 $\pm$ 5.30	<u>55.98</u> $\pm$ 3.66	<u>54.71</u> $\pm$ 4.41	<u>39.57</u> $\pm$ 5.62	82.56 $\pm$ 7.51	87.64 $\pm$ 2.61	81.90 $\pm$ 5.44	79.73 $\pm$ 6.18
CeFMC [19]	48.79 $\pm$ 4.62	47.75 $\pm$ 1.97	50.89 $\pm$ 3.73	34.15 $\pm$ 4.46	62.03 $\pm$ 0.41	51.90 $\pm$ 1.14	48.32 $\pm$ 1.66	42.30 $\pm$ 1.78
SFFedMC (Ours)	62.94 $\pm$ 0.97	55.54 $\pm$ 0.22	62.93 $\pm$ 0.29	48.43 $\pm$ 0.34	90.10 $\pm$ 0.02	80.79 $\pm$ 0.03	81.66 $\pm$ 0.04	<u>79.62</u> $\pm$ 0.04

Method	Mfeat				Multi-Fashion			
	ACC ($\uparrow$)	NMI ($\uparrow$)	F-score ($\uparrow$)	ARI ($\uparrow$)	ACC ($\uparrow$)	NMI ($\uparrow$)	F-score ($\uparrow$)	ARI ($\uparrow$)
FedMVL [16]	81.56 $\pm$ 7.31	79.87 $\pm$ 2.79	76.06 $\pm$ 5.21	73.28 $\pm$ 5.87	71.71 $\pm$ 4.58	80.34 $\pm$ 0.27	69.20 $\pm$ 2.75	65.27 $\pm$ 3.31
FedCMv [1]	69.94 $\pm$ 5.13	73.68 $\pm$ 2.62	66.38 $\pm$ 3.99	62.43 $\pm$ 4.51	76.66 $\pm$ 0.87	<u>81.35</u> $\pm$ 0.23	<u>72.62</u> $\pm$ 0.91	<u>69.36</u> $\pm$ 1.07
FedFCMv [1]	80.05 $\pm$ 8.89	77.56 $\pm$ 5.48	73.38 $\pm$ 8.34	70.35 $\pm$ 9.33	<u>76.68</u> $\pm$ 0.80	81.37 $\pm$ 0.22	72.90 $\pm$ 0.84	69.66 $\pm$ 0.98
FCUIF [25]	90.94 $\pm$ 0.53	84.32 $\pm$ 0.37	83.09 $\pm$ 0.91	81.20 $\pm$ 1.01	72.33 $\pm$ 1.68	75.74 $\pm$ 1.24	67.13 $\pm$ 1.82	63.21 $\pm$ 2.04
FMVFCMSP [8]	54.89 $\pm$ 3.40	55.07 $\pm$ 2.77	47.24 $\pm$ 3.54	40.96 $\pm$ 3.93	54.04 $\pm$ 3.13	60.65 $\pm$ 3.89	49.47 $\pm$ 3.40	42.69 $\pm$ 3.83
ESFMC [3]	<u>94.01</u> $\pm$ 0.93	<u>88.05</u> $\pm$ 1.73	<u>88.41</u> $\pm$ 1.78	<u>87.12</u> $\pm$ 1.98	73.60 $\pm$ 3.12	81.01 $\pm$ 3.87	70.23 $\pm$ 3.22	66.53 $\pm$ 3.51
CeFMC [19]	92.70 $\pm$ 0.14	85.00 $\pm$ 0.19	86.05 $\pm$ 0.25	84.50 $\pm$ 0.27	61.34 $\pm$ 0.22	65.31 $\pm$ 0.56	57.92 $\pm$ 0.53	52.57 $\pm$ 0.56
SFFedMC (Ours)	94.79 $\pm$ 0.22	88.81 $\pm$ 0.28	89.92 $\pm$ 0.41	88.80 $\pm$ 0.46	77.64 $\pm$ 0.53	77.45 $\pm$ 0.22	69.53 $\pm$ 0.41	65.88 $\pm$ 0.47

Method	MSRC-v1				ORL			
	ACC ($\uparrow$)	NMI ($\uparrow$)	F-score ($\uparrow$)	ARI ($\uparrow$)	ACC ($\uparrow$)	NMI ($\uparrow$)	F-score ($\uparrow$)	ARI ($\uparrow$)
FedMVL [16]	67.62 $\pm$ 5.20	59.09 $\pm$ 2.30	56.47 $\pm$ 4.03	49.07 $\pm$ 4.93	64.80 $\pm$ 2.43	81.38 $\pm$ 1.36	52.10 $\pm$ 2.83	50.83 $\pm$ 2.92
FedCMv [1]	69.71 $\pm$ 2.80	66.18 $\pm$ 1.85	61.25 $\pm$ 2.41	54.40 $\pm$ 2.97	57.95 $\pm$ 2.51	78.75 $\pm$ 1.77	47.51 $\pm$ 3.17	46.08 $\pm$ 3.29
FedFCMv [1]	75.90 $\pm$ 5.32	68.58 $\pm$ 3.47	64.97 $\pm$ 4.67	59.01 $\pm$ 5.57	61.25 $\pm$ 2.21	80.25 $\pm$ 1.85	49.24 $\pm$ 2.79	47.87 $\pm$ 2.88
FCUIF [25]	<u>86.19</u> $\pm$ 1.44	<u>77.34</u> $\pm$ 1.45	<u>76.40</u> $\pm$ 1.70	<u>72.48</u> $\pm$ 1.99	58.45 $\pm$ 2.49	76.54 $\pm$ 1.07	45.59 $\pm$ 2.47	44.23 $\pm$ 2.54
FMVFCMSP [8]	56.19 $\pm$ 2.63	48.19 $\pm$ 3.62	43.85 $\pm$ 2.69	33.32 $\pm$ 2.87	49.00 $\pm$ 3.99	71.23 $\pm$ 1.94	35.51 $\pm$ 4.11	33.80 $\pm$ 4.29
ESFMC [3]	73.90 $\pm$ 2.12	64.74 $\pm$ 3.25	61.28 $\pm$ 2.80	54.75 $\pm$ 3.26	70.20 $\pm$ 2.48	84.60 $\pm$ 1.15	59.72 $\pm$ 2.53	58.70 $\pm$ 2.61
CeFMC [19]	76.67 $\pm$ 2.74	66.87 $\pm$ 3.36	65.78 $\pm$ 3.14	59.79 $\pm$ 3.78	<u>76.35</u> $\pm$ 2.08	88.74 $\pm$ 1.08	69.02 $\pm$ 2.45	68.25 $\pm$ 2.52
SFFedMC (Ours)	89.14 $\pm$ 0.70	81.91 $\pm$ 1.07	79.94 $\pm$ 1.01	76.63 $\pm$ 1.17	84.25 $\pm$ 1.15	92.50 $\pm$ 0.67	77.60 $\pm$ 1.46	77.04 $\pm$ 1.49

Method	WebKB				Yale			
	ACC ($\uparrow$)	NMI ($\uparrow$)	F-score ($\uparrow$)	ARI ($\uparrow$)	ACC ($\uparrow$)	NMI ($\uparrow$)	F-score ($\uparrow$)	ARI ($\uparrow$)
FedMVL [16]	72.32 $\pm$ 9.38	46.23 $\pm$ 8.82	68.95 $\pm$ 7.98	<u>45.91</u> $\pm$ 12.95	62.79 $\pm$ 2.67	67.16 $\pm$ 2.63	50.09 $\pm$ 3.08	46.67 $\pm$ 3.30
FedCMv [1]	71.33 $\pm$ 2.19	29.53 $\pm$ 1.77	63.38 $\pm$ 1.38	36.49 $\pm$ 0.57	46.18 $\pm$ 7.97	57.24 $\pm$ 7.09	35.49 $\pm$ 8.36	30.09 $\pm$ 9.55
FedFCMv [1]	71.43 $\pm$ 0.44	27.60 $\pm$ 0.62	63.15 $\pm$ 0.87	35.83 $\pm$ 1.21	50.30 $\pm$ 5.78	58.84 $\pm$ 5.58	37.77 $\pm$ 5.81	32.99 $\pm$ 6.32
FCUIF [25]	50.05 $\pm$ 5.51	23.27 $\pm$ 1.73	46.51 $\pm$ 2.01	21.59 $\pm$ 2.77	58.42 $\pm$ 3.67	60.56 $\pm$ 2.75	41.98 $\pm$ 3.49	38.03 $\pm$ 3.73
FMVFCMSP [8]	66.21 $\pm$ 4.41	26.29 $\pm$ 7.45	60.69 $\pm$ 3.42	26.77 $\pm$ 12.96	41.09 $\pm$ 5.11	49.73 $\pm$ 3.62	28.01 $\pm$ 4.82	22.37 $\pm$ 5.50
ESFMC [3]	51.33 $\pm$ 10.85	10.12 $\pm$ 5.64	46.22 $\pm$ 8.24	11.20 $\pm$ 8.53	61.82 $\pm$ 1.58	64.44 $\pm$ 2.78	45.55 $\pm$ 3.36	41.74 $\pm$ 3.59
CeFMC [19]	63.35 $\pm$ 8.01	16.99 $\pm$ 10.82	57.87 $\pm$ 5.39	22.52 $\pm$ 14.56	64.36 $\pm$ 3.44	68.19 $\pm$ 4.63	52.86 $\pm$ 5.36	49.61 $\pm$ 5.73
SFFedMC (Ours)	79.90 $\pm$ 0.57	48.42 $\pm$ 2.26	75.27 $\pm$ 1.48	56.01 $\pm$ 3.25	74.67 $\pm$ 0.89	76.71 $\pm$ 1.15	63.25 $\pm$ 1.68	60.78 $\pm$ 1.80

4.4 Convergence Analysis

Figure 5 illustrates the convergence behavior on the Mfeat dataset. The objective function (left) stabilizes within 3 iterations and remains steady thereafter. The relative change (right) drops below the tolerance threshold (10^{-5}) rapidly, demonstrating fast convergence.

This efficiency can be attributed to the warm initialization strategy and the closed-form updates derived in Section 3.3, which ensure each BCA-CR step monotonically decreases the objective. This pattern is consistent across all datasets (see Appendix F.4 for details), confirming the efficiency and stability of the BCA-CR optimization.

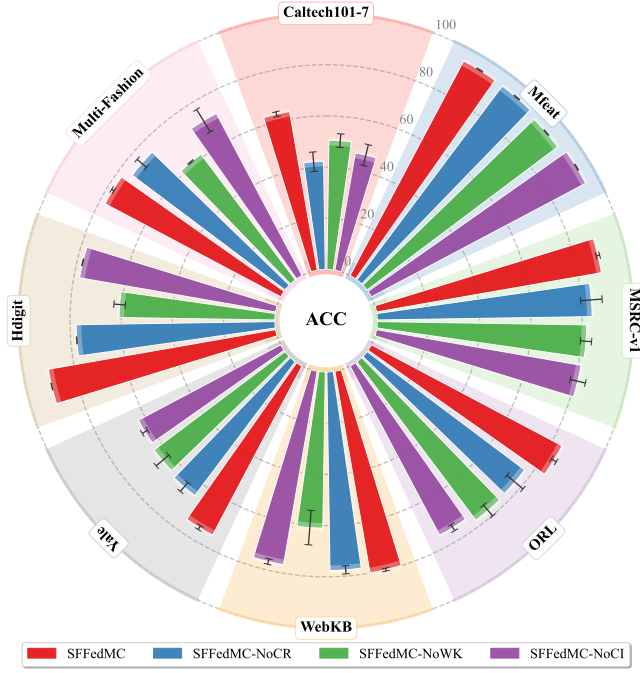


Figure 4: Ablation study on the ACC metric. The full SFFedMC method achieves the best performance, validating the necessity of each component.

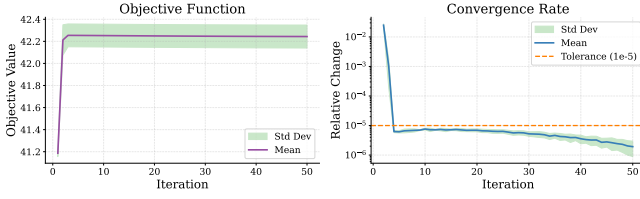


Figure 5: Convergence analysis on the Mfeat dataset. Our method converges within 3 iterations and maintains stability.

4.5 Parameter Sensitivity Analysis

Figure 6 examines sensitivity to λ , β , and γ on ACC. Overall, SFFedMC exhibits robust performance across a wide range of hyperparameter values, reflecting the stability of the proposed optimization framework. Only WebKB shows slight sensitivity: performance degrades when $\lambda < 0.1$, $\beta > 30$, or $\gamma > 0.3$. This suggests that extreme hyperparameter settings may disrupt the balance between local clustering, consensus integration, and causal regularization. In practice, we find that $\lambda \in [0.1, 100]$, $\beta \in [0.01, 10]$, and $\gamma \in [0.0001, 0.1]$ consistently yield strong performance across all evaluated datasets, providing a practical guideline for hyperparameter selection. Results for other metrics are provided in [Appendix F.5](#).

4.6 Visualization Analysis

Figure 7 visualizes the learned representations and causal weights on the Mfeat dataset. The t-SNE plot (top) shows well-separated clusters in the learned embedding space, validating the quality of the

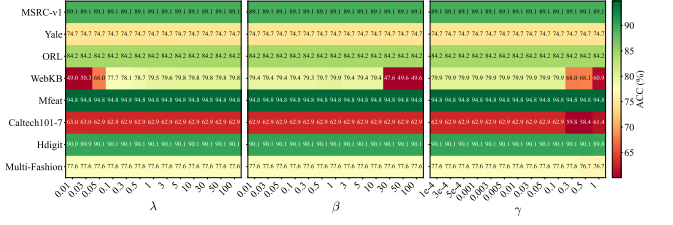


Figure 6: Parameter sensitivity analysis on the ACC metric.

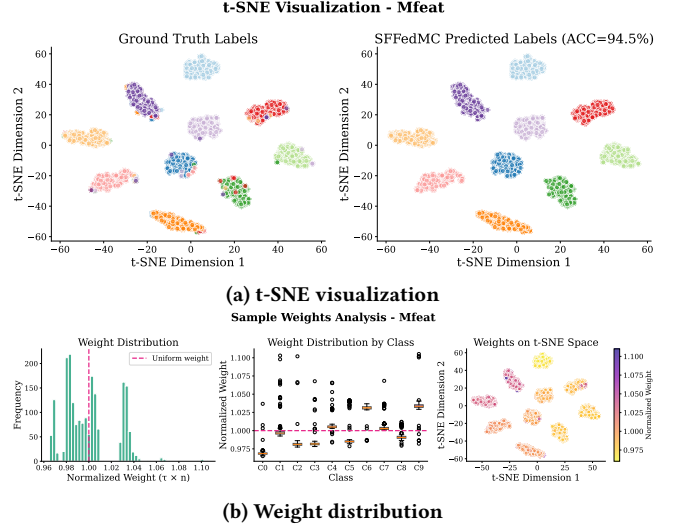


Figure 7: Visualization analysis on the Mfeat dataset.

consensus representation. The weight distribution (bottom) reveals that SFFedMC assigns higher weights to samples with more informative features, effectively downweighting confounder-affected samples whose feature-label correlations are spurious rather than causal. The bimodal nature of the weight distribution further suggests that the confounder balancing mechanism successfully identifies two distinct groups of samples—those well-explained by causal features and those corrupted by confounders. Additional visualizations are provided in [Appendix F.6](#).

5 Conclusions

In this paper, we introduced SFFedMC, a novel federated multi-view clustering framework that addresses spurious correlations via causal inference. Grounded in a structural causal model, SFFedMC identifies backdoor paths induced by hidden confounders and eliminates their influence using a privacy-preserving confounder-balancing mechanism, thereby distinguishing causal features from spurious ones without exchanging raw data. Extensive experiments on eight real-world datasets demonstrate that SFFedMC consistently outperforms state-of-the-art baselines across diverse datasets while maintaining computational efficiency. Future work includes extending to dynamic federated environments with evolving data distributions, handling incomplete multi-view data, and leveraging large language model (LLM) priors to guide causal feature learning [29].

Acknowledgments

This work was supported by the National Natural Science Foundation of China (Grant Nos. 62506174, 62376087, 62306008, 62502233), the Basic Research Program of Jiangsu (Grant Nos. BK20250656, BK20250650), the Natural Science Research Start-up Foundation of Recruiting Talents of Nanjing University of Posts and Telecommunications (Grant Nos. NY225023, NY224061), the Natural Science Foundation of the Jiangsu Higher Education Institutions of China (Grant Nos. 25KJB520036, 25KJB520033), the Ministry of Education, Singapore, under its Academic Research Fund Tier 1 (RG101/24), and the Key Project of Anhui Province's Science and Technology Innovation Tackle Plan (Grant No. 202423k09020040).

References

- [1] Jose Luis Corcuera Barcena, Francesco Marcelloni, Alessandro Renda, Alessio Bechini, and Pietro Ducange. 2024. Federated c-means and fuzzy c-means clustering algorithms for horizontally and vertically partitioned data. *IEEE Transactions on Artificial Intelligence* 5, 12 (2024), 6426–6441.
- [2] Alain Berline and Christine Thomas-Agnan. 2011. *Reproducing kernel Hilbert spaces in probability and statistics*. Springer Science & Business Media.
- [3] Xinyue Chen, Jinfeng Peng, Yuhao Li, Xiaorong Pu, Yang Yang, and Yazhou Ren. 2025. An Effective and Secure Federated Multi-View Clustering Method with Information-Theoretic Perspective. In *International Conference on Machine Learning*. PMLR, 8871–8889.
- [4] Xinyue Chen, Yazhou Ren, Jie Xu, Fangfei Lin, Xiaorong Pu, and Yang Yang. 2024. Bridging gaps: Federated multi-view clustering in heterogeneous hybrid views. *Advances in Neural Information Processing Systems* 37 (2024), 37020–37049.
- [5] Xinyue Chen, Jie Xu, Yazhou Ren, Xiaorong Pu, Ce Zhu, Xiaofeng Zhu, Zhifeng Hao, and Lifang He. 2023. Federated deep multi-view clustering with global self-supervision. In *Proceedings of the ACM International Conference on Multimedia*. 3498–3506.
- [6] John Duchi, Shai Shalev-Shwartz, Yoram Singer, and Tushar Chandra. 2008. Efficient projections onto the l_1 -ball for learning in high dimensions. In *Proceedings of the International Conference on Machine Learning*. 272–279.
- [7] Wei Feng, Danting Liu, Qianqian Wang, Wenqi Liang, and Zheng Yan. 2025. Scalable Federated One-Step Multi-View Clustering with Tensorized Regularization. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 39. 16586–16594.
- [8] Wei Feng, Zhenwei Wu, Qianqian Wang, Bo Dong, and Quanxue Gao. 2024. Federated Fuzzy C-means with Schatten-p Norm Minimization. In *Proceedings of the ACM International Conference on Multimedia*. 9407–9416.
- [9] Wei Feng, Zhenwei Wu, Qianqian Wang, Bo Dong, Zhiqiang Tao, and Quanxue Gao. 2024. Efficient federated multi-view clustering with integrated matrix factorization and K-means. In *Proceedings of the International Joint Conference on Artificial Intelligence*. 3971–3979.
- [10] Wei Feng, Zhenwei Wu, Qianqian Wang, Bo Dong, Zhiqiang Tao, and Quanxue Gao. 2024. Federated multi-view clustering via tensor factorization. In *Proceedings of the International Joint Conference on Artificial Intelligence*. 3962–3970.
- [11] Xianjie Guo, Kui Yu, Lizhen Cui, Han Yu, and Xiaoxiao Li. 2025. Federated causally invariant feature learning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 39. 16978–16986.
- [12] Xianjie Guo, Kui Yu, Lin Liu, and Jiuyong Li. 2024. FedCSL: A scalable and accurate approach to federated causal structure learning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 38. 12235–12243.
- [13] Xianjie Guo, Kui Yu, Lin Liu, Jiuyong Li, Jiye Liang, Fuyuan Cao, and Xindong Wu. 2024. Progressive skeleton learning for effective local-to-global causal structure learning. *IEEE Transactions on Knowledge and Data Engineering* 36, 12 (2024), 9065–9079.
- [14] Xianjie Guo, Kui Yu, Hao Wang, Lizhen Cui, Han Yu, and Xiaoxiao Li. 2024. Sample quality heterogeneity-aware federated causal discovery through adaptive variable space selection. In *Proceedings of the International Joint Conference on Artificial Intelligence*. 4071–4079.
- [15] Xingchen Hu, Jindong Qin, Yinghua Shen, Witold Pedrycz, Xinwang Liu, and Jiyuan Liu. 2024. An Efficient Federated Multiview Fuzzy C-Means Clustering Method. *IEEE Transactions on Fuzzy Systems* 32, 4 (2024), 1886–1899.
- [16] Shudong Huang, Wei Shi, Zenglin Xu, Ivor W Tsang, and Jiancheng Lv. 2022. Efficient federated multi-view learning. *Pattern Recognition* 131 (2022), 108817.
- [17] Kun Kuang, Peng Cui, Bo Li, Meng Jiang, and Shiqiang Yang. 2017. Estimating treatment effect in the wild via differentiated confounder balancing. In *Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. 265–274.
- [18] Atsutoshi Kumagai and Tomoharu Iwata. 2019. Unsupervised domain adaptation by matching distributions based on the maximum mean discrepancy via unilateral transformations. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 33. 4106–4113.
- [19] Jiyuan Liu, Xinwang Liu, Siqi Wang, Xinhang Wan, Dongsheng Li, Kai Lu, and Kunlun He. 2026. Communication-Efficient Federated Multi-View Clustering. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 48, 1 (2026), 17–32.
- [20] Suyuan Liu, Siwei Wang, Ke Liang, Junpu Zhang, Zhibin Dong, Tianrui Liu, En Zhu, Kunlun He, and Xinwang Liu. 2024. Alleviate anchor-shift: Explore blind spots with cross-view reconstruction for incomplete multi-view clustering. *Advances in Neural Information Processing Systems* 37 (2024), 87509–87531.
- [21] Suyuan Liu, Hao Yu, Hao Tan, Ke Liang, Siwei Wang, Shengyu Yu, En Zhu, and Xinwang Liu. 2025. Efficient Federated Incomplete Multi-View Clustering. In *Proceedings of the International Conference on Machine Learning*. PMLR, 40100–40114.
- [22] Suyuan Liu, Lijun Zhang, Siwei Wang, Miaomiao Li, Xueling Zhu, and Xinwang Liu. 2026. Towards Scalable Multi-View Clustering without Predefined Clusters via Inter-Anchor Graph Learning. *IEEE Transactions on Knowledge and Data Engineering* 38, 8 (2026), 5453–5465. doi:10.1109/TKDE.2026.3699654
- [23] Abdelmalik Moujahid and Fadi Dornaika. 2025. Advanced unsupervised learning: a comprehensive overview of multi-view clustering techniques. *Artificial Intelligence Review* 58, 8 (2025), 234.
- [24] Judea Pearl. 2009. *Causality*. Cambridge University Press.
- [25] Yazhou Ren, Xinyue Chen, Jie Xu, Jingyu Pu, Yonghao Huang, Xiaorong Pu, Ce Zhu, Xiaofeng Zhu, Zhifeng Hao, and Lifang He. 2024. A novel federated multi-view clustering method for unaligned and incomplete data fusion. *Information Fusion* 108 (2024), 102357.
- [26] Xiaoli Tang and Han Yu. 2025. Reputation-aware revenue allocation for auction-based federated learning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, Vol. 39. 20832–20840.
- [27] Xiaoli Tang, Han Yu, and Xiaoxiao Li. 2026. A Reinforcement Learning-based Bidding Strategy for Data Consumers in Auction-based Federated Learning. *Advances in Neural Information Processing Systems* 38 (2026), 82666–82684.
- [28] Xiaoli Tang, Han Yu, Zengxiang Li, and Xiaoxiao Li. 2025. Multi-Session Budget Optimization for Forward Auction-based Federated Learning. In *Proceedings of the International Conference on Machine Learning*. PMLR, 59005–59016.
- [29] Xingyu Wu, Yu Zhou, and Kay Chen Tan. 2026. Building LLMs Like LEGO: Two-dimensional Architecture Reassembly of Large Language Models. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. 44939–44956.
- [30] Xueming Yan, Ziqi Wang, and Yaochu Jin. 2024. Federated Incomplete Multi-view Clustering with Heterogeneous Graph Neural Networks. In *International Workshop on Trustworthy Federated Learning*. Springer, 61–76.
- [31] Miin-Shen Yang and Kristina P Sinaga. 2025. Federated Multi-View K-Means Clustering. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 47, 4 (2025), 2446–2459.
- [32] Xihong Yang, Siwei Wang, Jiaqi Jin, Fangdi Wang, Tianrui Liu, Yueming Jin, Xinwang Liu, En Zhu, and Kunlun He. 2025. Generalized Deep Multi-view Clustering via Causal Learning with Partially Aligned Cross-view Correspondence. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*. 1990–1999.
- [33] Kui Yu, Xianjie Guo, Lin Liu, Jiuyong Li, Hao Wang, Zhaolong Ling, and Xindong Wu. 2020. Causality-based feature selection: Methods and evaluations. *ACM Computing Surveys (CSUR)* 53, 5 (2020), 1–36.
- [34] Shengju Yu, Zhibin Dong, Siwei Wang, Suyuan Liu, Ke Liang, Xinwang Liu, Yue Liu, and Yi Zhang. 2025. From Spectrum-free towards Baseline-view-free: Double-track Proximity Driven Multi-view Clustering. In *Proceedings of the International Conference on Machine Learning*. PMLR, 72895–72924.
- [35] Shengju Yu, Suyuan Liu, Wenhao Shao, Siwei Wang, Ke Liang, Xihong Yang, Tiejun Li, and Xinwang Liu. 2026. Plug-and-Play Incomplete Multi-View Clustering via Janus-Faced Affinity Learning with Topology Harmonization. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 3594–3603.
- [36] Shengju Yu, Pei Zhang, Siwei Wang, Suyuan Liu, Xinhang Wan, Zhibin Dong, Tiejun Li, and Xinwang Liu. 2026. Bit-swapping Oriented Twin-memory Multi-view Clustering in Lifelong Incomplete Scenarios. *Advances in Neural Information Processing Systems* 38 (2026), 69987–70024.
- [37] Lihua Zhou, Guowang Du, Kevin Lue, Lihong Wang, and Jingwei Du. 2024. A survey and an empirical evaluation of multi-view clustering approaches. *Comput. Surveys* 56, 7 (2024), 1–38.